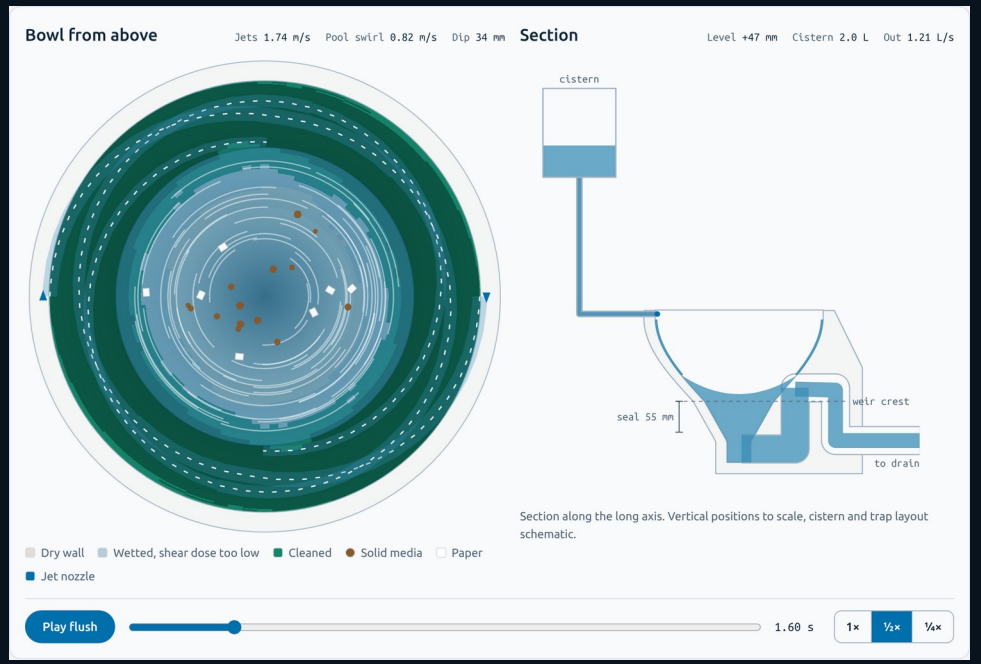
Tangential Jet Flush Model 2.0

A fast physics model of a rimless toilet flush, checked against CFD, with AI in the engineering loop



Flush playback from the page: films on the wall, pool swirl, waste. Washdown preset.

20–40 ms

per flush with the fast model, in the browser

≈ 2 hours

for 3.2 s of the same flush in OpenFOAM CFD, on 12 cores

4.5 vs 12

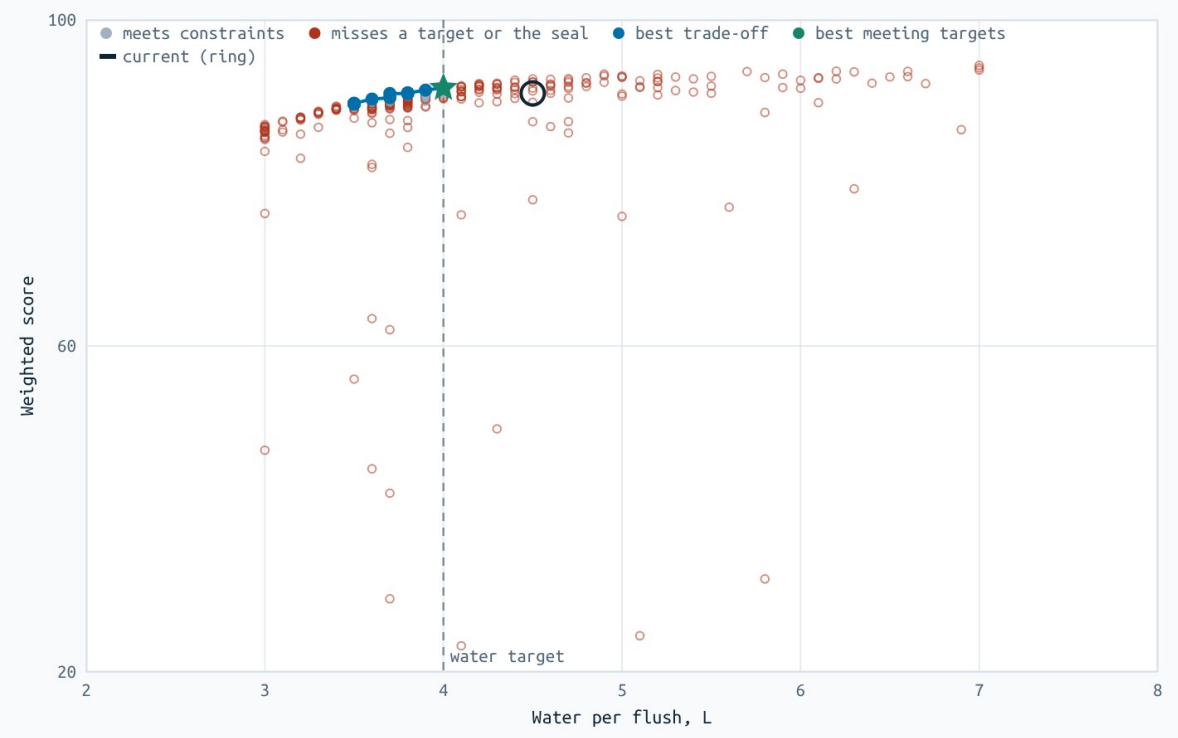
expensive runs to reach the targets, AI-assisted vs one change at a time (median)

Try it live fintlp.github.io/jet-flush-model

Reduced-order model, coefficients not yet calibrated: results show trends between designs, not validated values. The CFD is a second simulation, not a measurement.

1 The fast model: one flush in milliseconds

Change the bowl or the jets and see cleaning, waste removal, water and noise at once; the AI lab searches for better designs.



AI lab: each dot is one simulated flush; the curve is the best trade-off between score and water.

From the guided demo, part 1

86 % → 79 %

cleaning when the bowl turns oval, jets at the ends of the long axis

→ 88 %

after moving the jets to the long sides

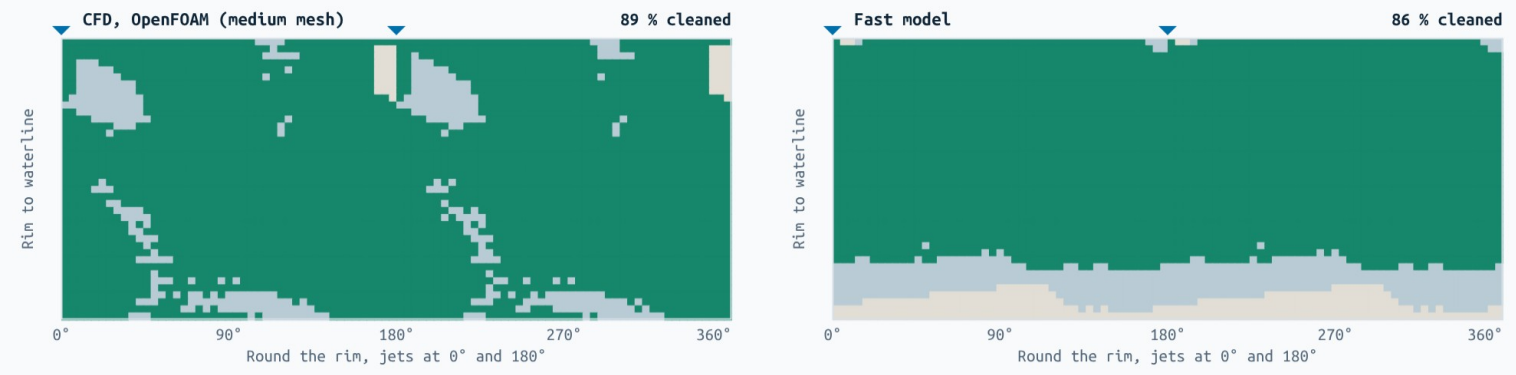
89 % at 4.0 L

AI lab design: 0.5 L less water than the oval start, −0.5 dB, no louder

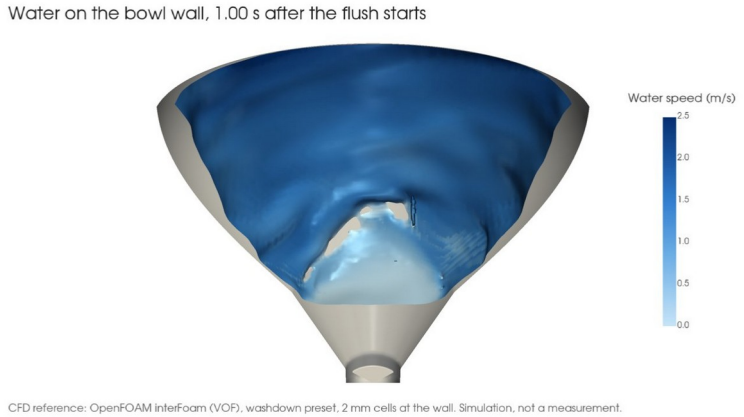
Illustrative estimates from the uncalibrated model; noise is a relative, experimental indicator. AI suggestions are candidates for a CFD or rig check, not final answers.

2 Checked against CFD

The same bowl and jets in OpenFOAM: about 2 hours on 12 cores for 3.2 s of flush, against about 30 ms for the fast model.



What gets cleaned, wall unrolled: CFD left, fast model right (green cleaned, grey wetted only).



CFD water surface at 1.0 s, front half removed.

89 % vs 86 %

of the wall cleaned, CFD vs fast model: close totals, different patterns

0.14 s vs 0.54 s

first water in the pool, CFD vs fast model: in the model it arrives later and all at once

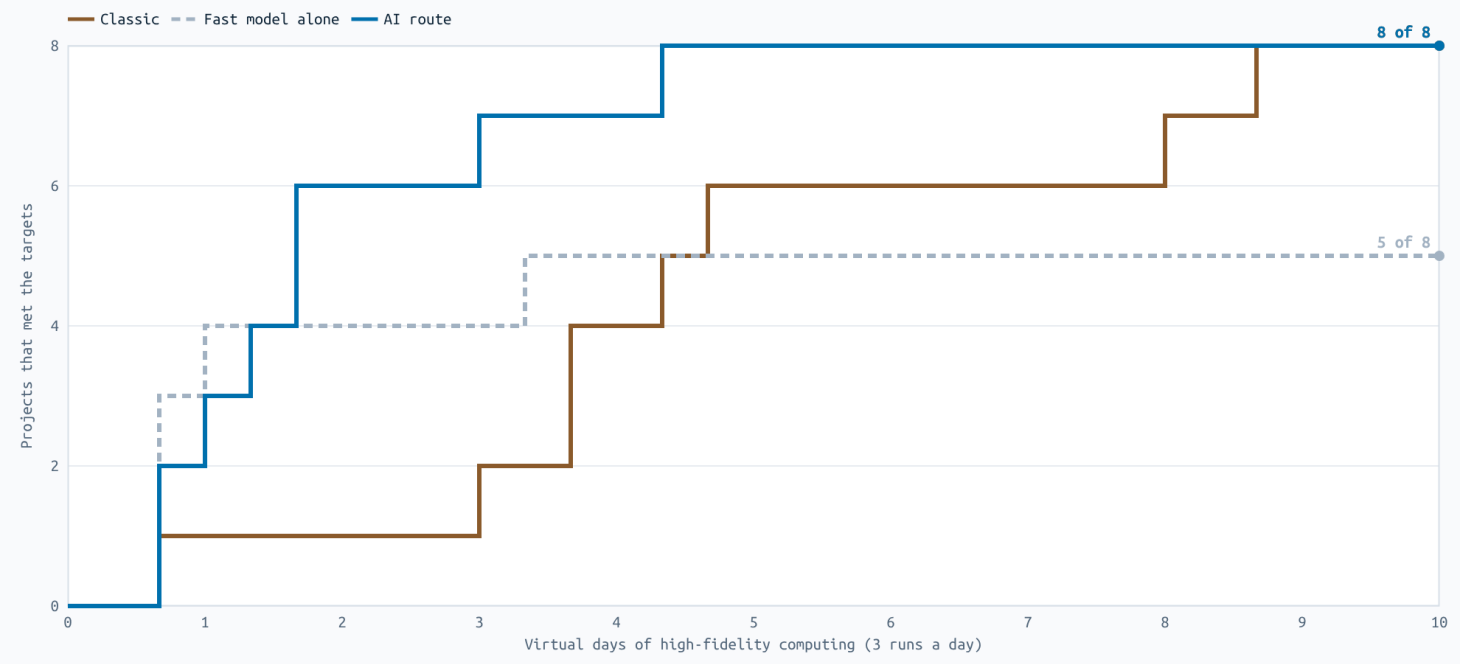
92 %

of the wall gets the same cleaned verdict on two CFD meshes (2 and 3 mm cells)

A second simulation, not a measurement or a validation. The trap outflow is taken from the model; after 1.3 s the CFD outlet draws air and its bowl holds more water. Two meshes show a sensitivity, not mesh independence.

3 AI in the engineering loop

Eight design projects, each run of the expensive reference costs 8 hours: how many runs until the targets are met?



Projects that met the targets over virtual days, for the three routes.

4.5 vs 12

expensive runs per project (median): fast model + learned correction vs one change at a time

43 vs 110

runs for all eight projects

3 of 8

projects the fast model alone never finished: it cannot learn from the expensive runs

The expensive runs come from a synthetic reference (the model with other coefficients and extra effects), labelled as such on the page. It shows the workflow, not real accuracy.

One design loop, two clocks

The median project of the challenge at one working day per expensive iteration, a figure Geberit reported for its bowl development (an assumption here).

One change at a time



12 days

AI-assisted



4.5 days

≈ 3 months

of iterations saved over the eight projects

110 → 43

expensive iterations in total

Also in the AI lab, all in the browser

Design optimiser: about 540 designs in about 10 s, checked at full resolution

Neural surrogate: a few thousand times faster than the physics model

Hybrid calibration: fit plus a learned correction, tested on unseen designs

Noise calibration: fits the noise model to phone recordings

Physics-informed network: finds a model coefficient from a level record

Field surrogate: whole cleaning maps from expensive runs, shown honestly

Iteration counts measured in the challenge (checks/challenge.js); one day per iteration is an assumption from the Geberit report by CADFEM. Sources on the page.

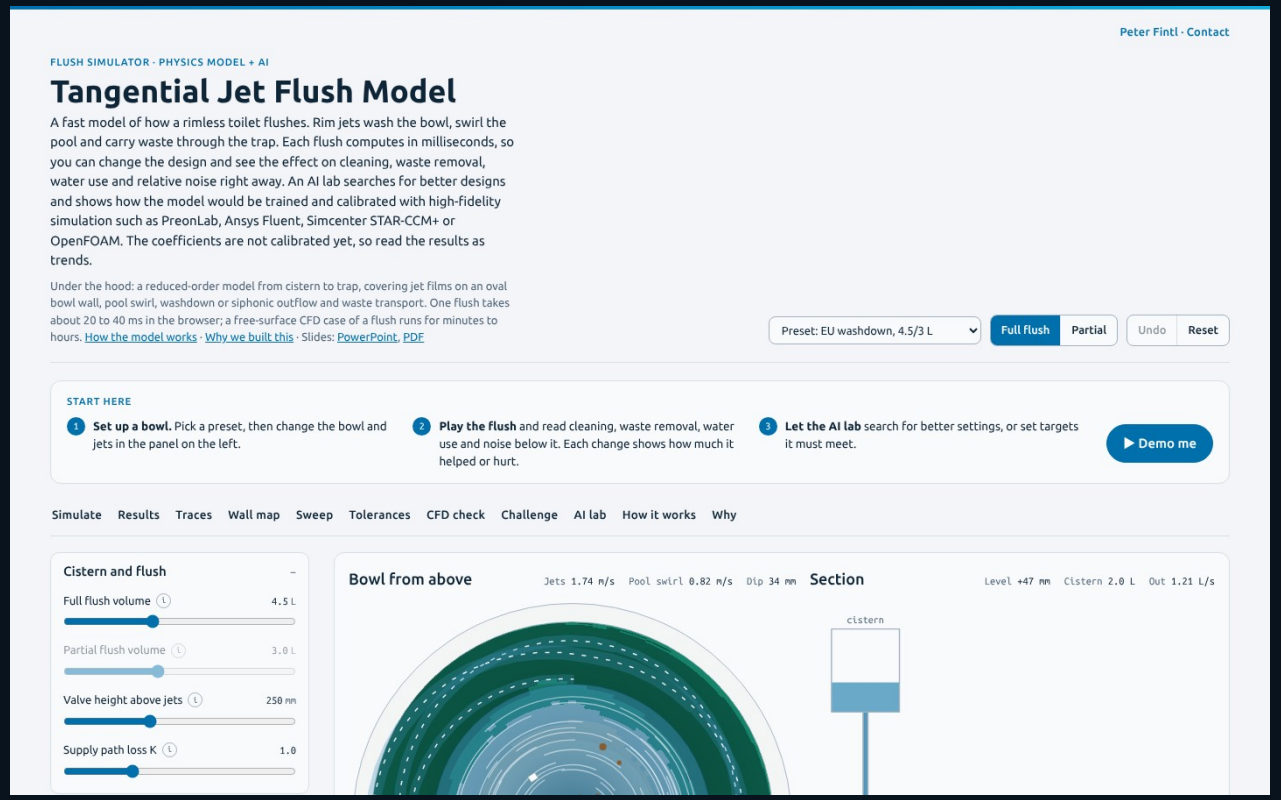
See the whole story

The guided story drives the live page in three parts. Each part can be started on its own.

1. **The fast model**
   fintlp.github.io/jet-flush-model/#demo

2. **Checked against CFD**
   fintlp.github.io/jet-flush-model/#cfd-tour

3. **AI in the engineering loop**
   fintlp.github.io/jet-flush-model/#challenge-tour



Recorded from the live page (v2.0.0) on 21 October 2026: all three parts, at reading pace.

Physics model, uncalibrated: trends between designs, not validated values. Synthetic references are labelled on the page; the CFD is a second simulation, not a measurement.